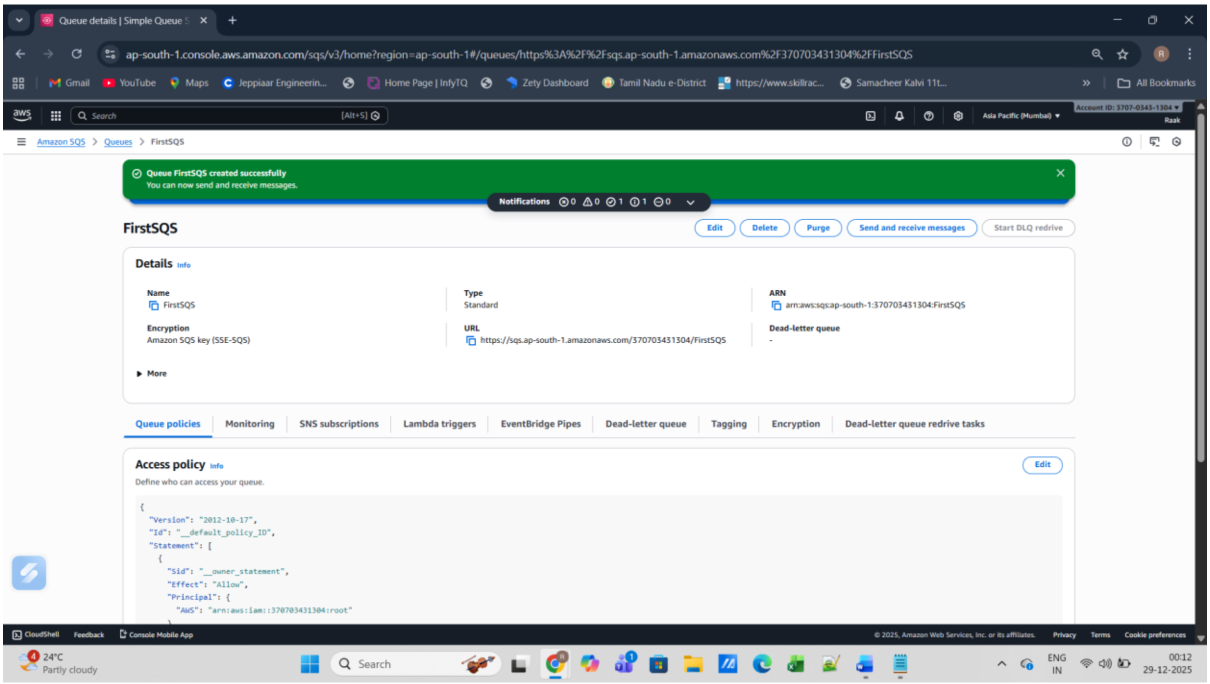
# WEEK 7 ASSIGNMENT: Serverless Workflow Automation Project

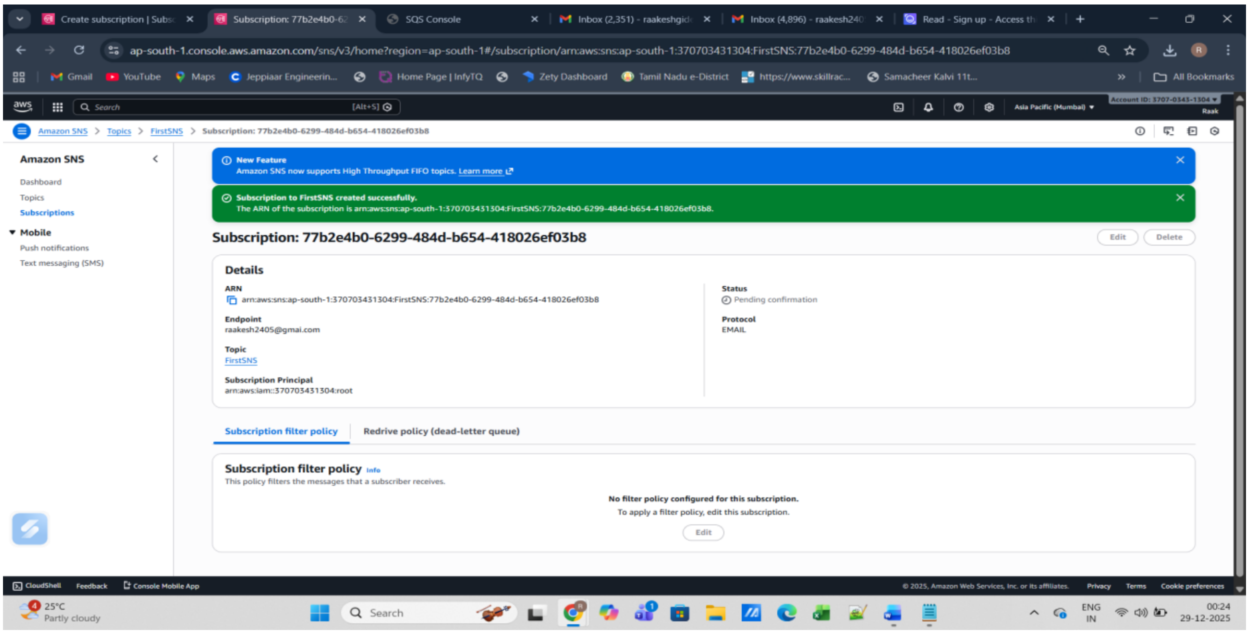
## Objective:

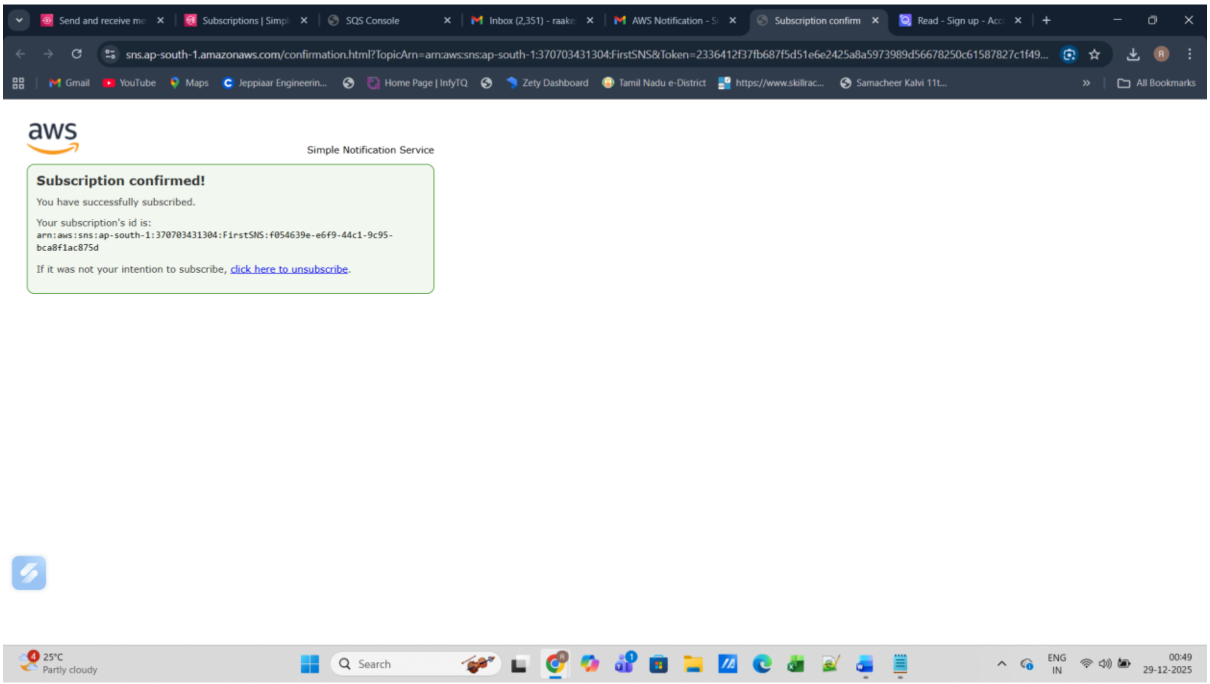
The objective of this project is to design and implement an event-driven serverless architecture using AWS services. The system processes messages sent to an SQS queue using AWS Lambda, sends email notifications via SNS upon successful execution, and monitors errors using CloudWatch alarms.

SQS created:

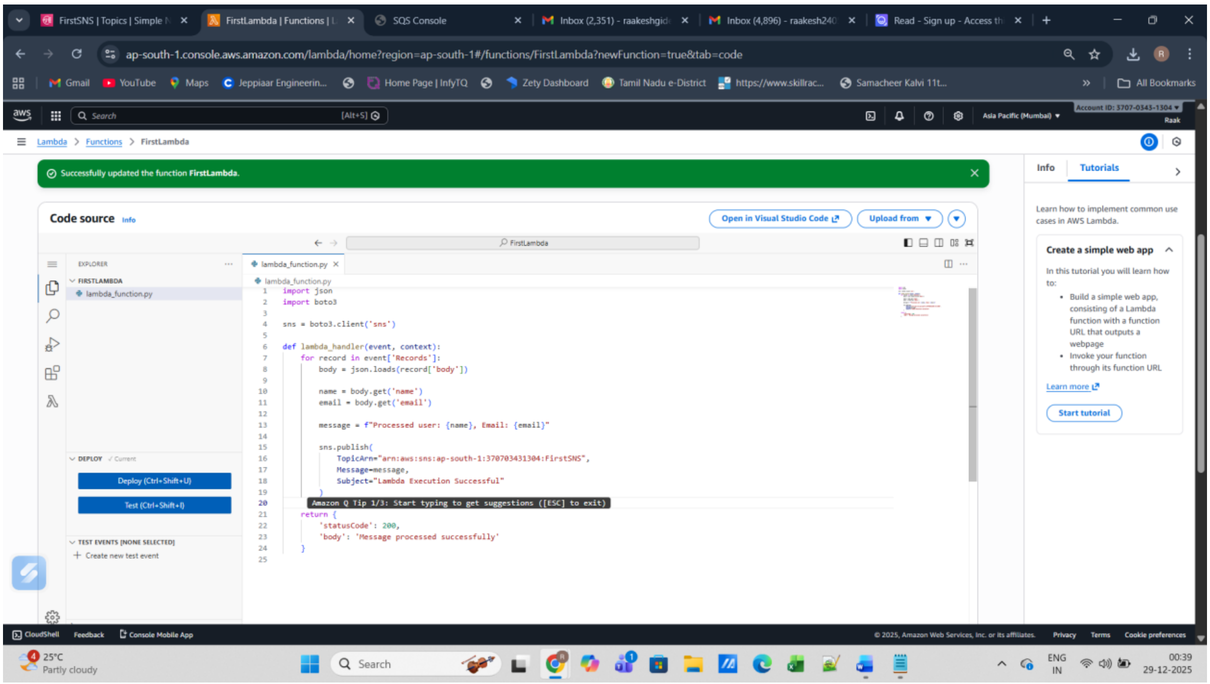


SNS configured:

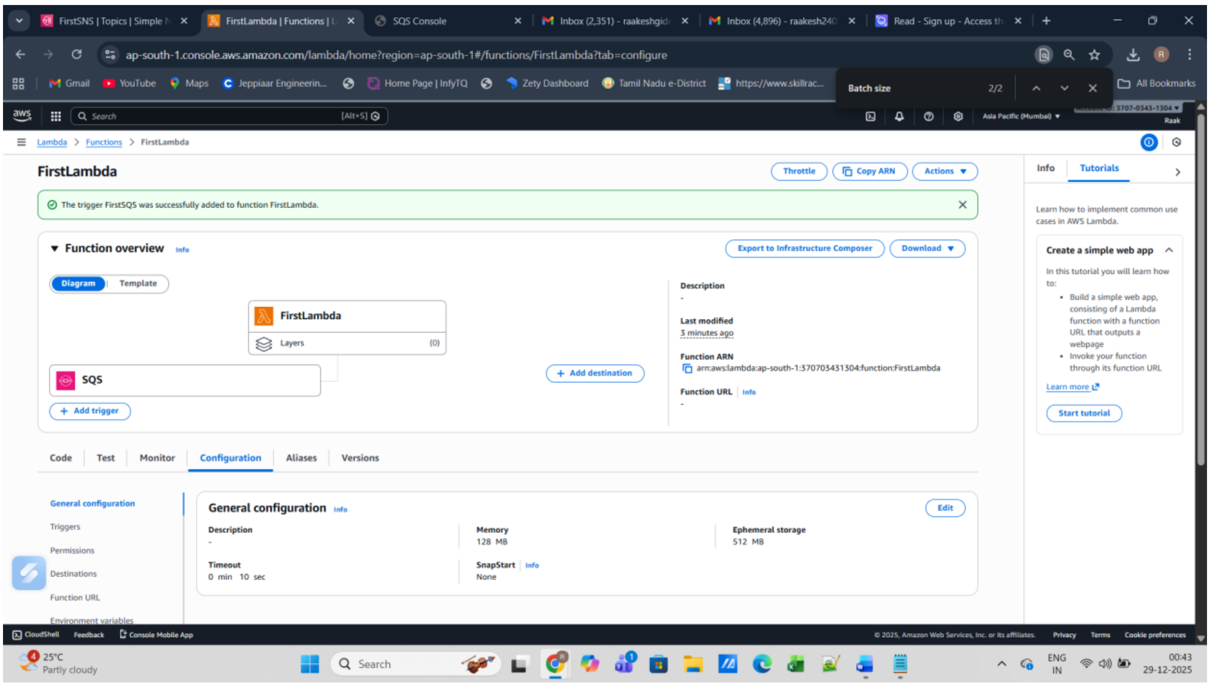


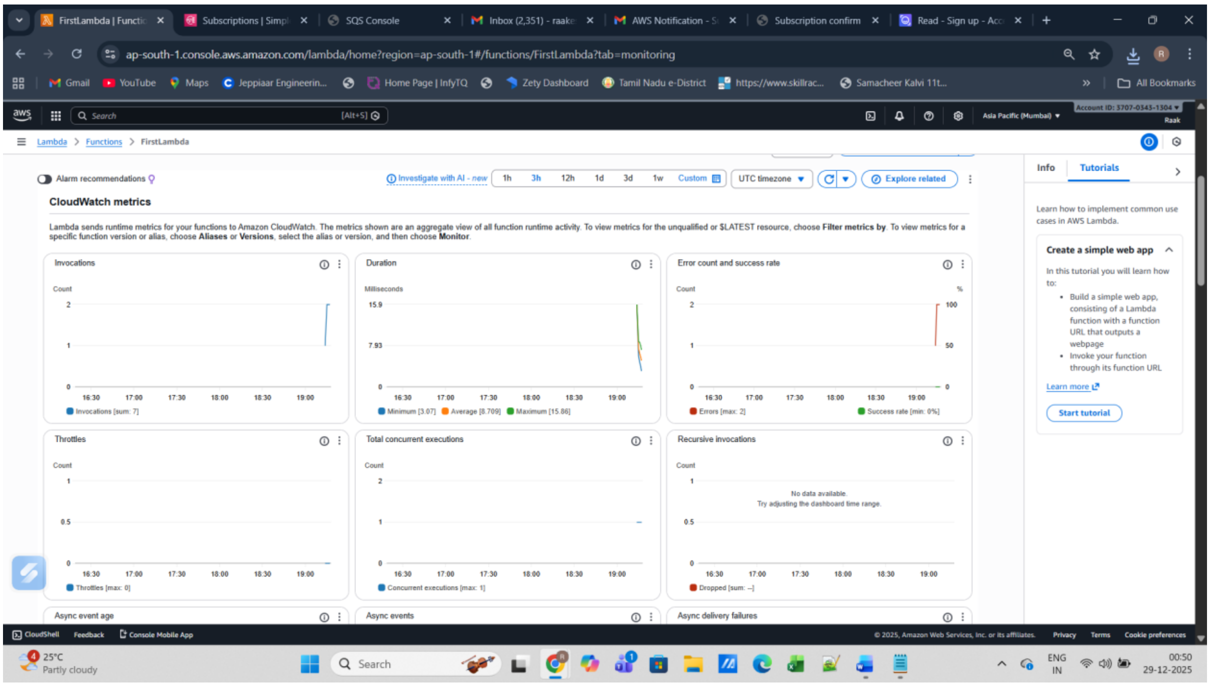



## Lambda configured:

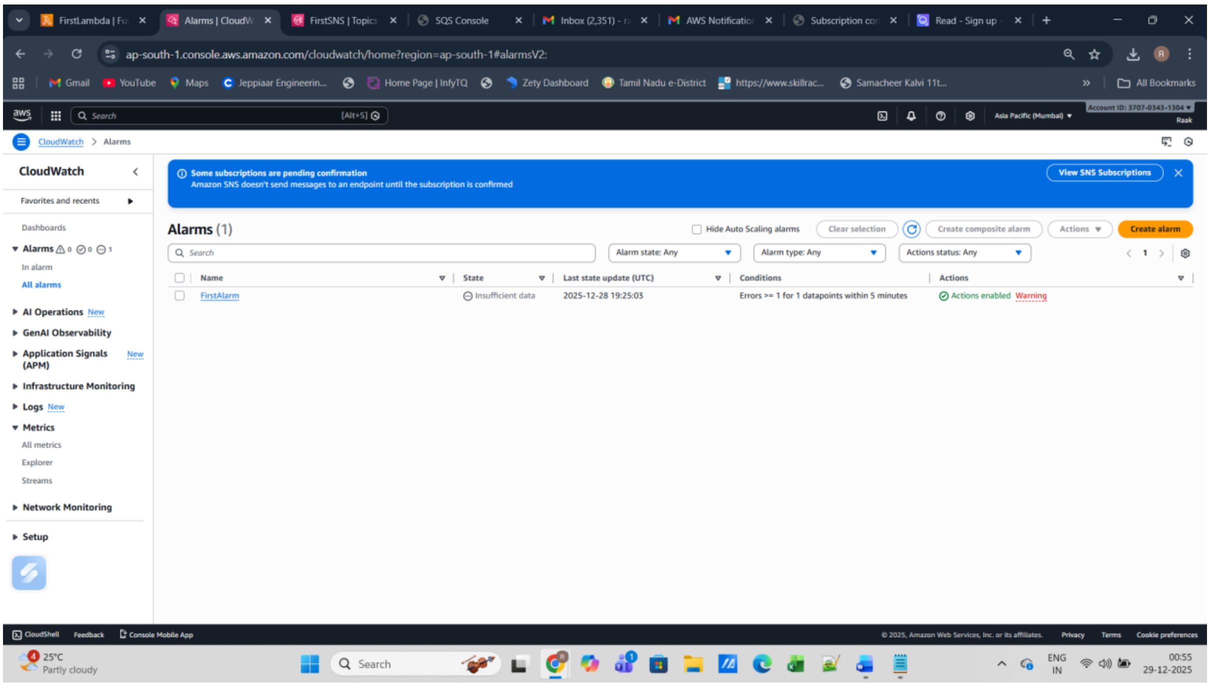


## SQS triggered through lambda:






## Alarm created in association with cloudwatch:



**Cloud formation YAM template:**

```yaml
AWSTemplateFormatVersion: '2010-09-09'
Description: Week 7 - Lambda with SQS and SNS
Resources:
 MySQSQueue:
  Type: AWS::SQS::Queue
 MySNSTopic:
  Type: AWS::SNS::Topic
 MyLambdaRole:
  Type: AWS::IAM::Role
  Properties:
   AssumeRolePolicyDocument:
    Version: '2012-10-17'
    Statement:
     - Effect: Allow
       Principal:
        Service: lambda.amazonaws.com
       Action: sts:AssumeRole
   ManagedPolicyArns:
    - arn:aws:iam::aws:policy/AmazonSQSFullAccess
    - arn:aws:iam::aws:policy/AmazonSNSFullAccess
    - arn:aws:iam::aws:policy/CloudWatchLogsFullAccess
 MyLambdaFunction:
  Type: AWS::Lambda::Function
  Properties:
   Runtime: python3.9
   Handler: index.lambda_handler
   Role: !GetAtt MyLambdaRole.Arn
   Code:
```

```
ZipFile: |
    def lambda_handler(event, context):
      print("Lambda triggered from SQS")            print("Lambda triggered from SQS")
```